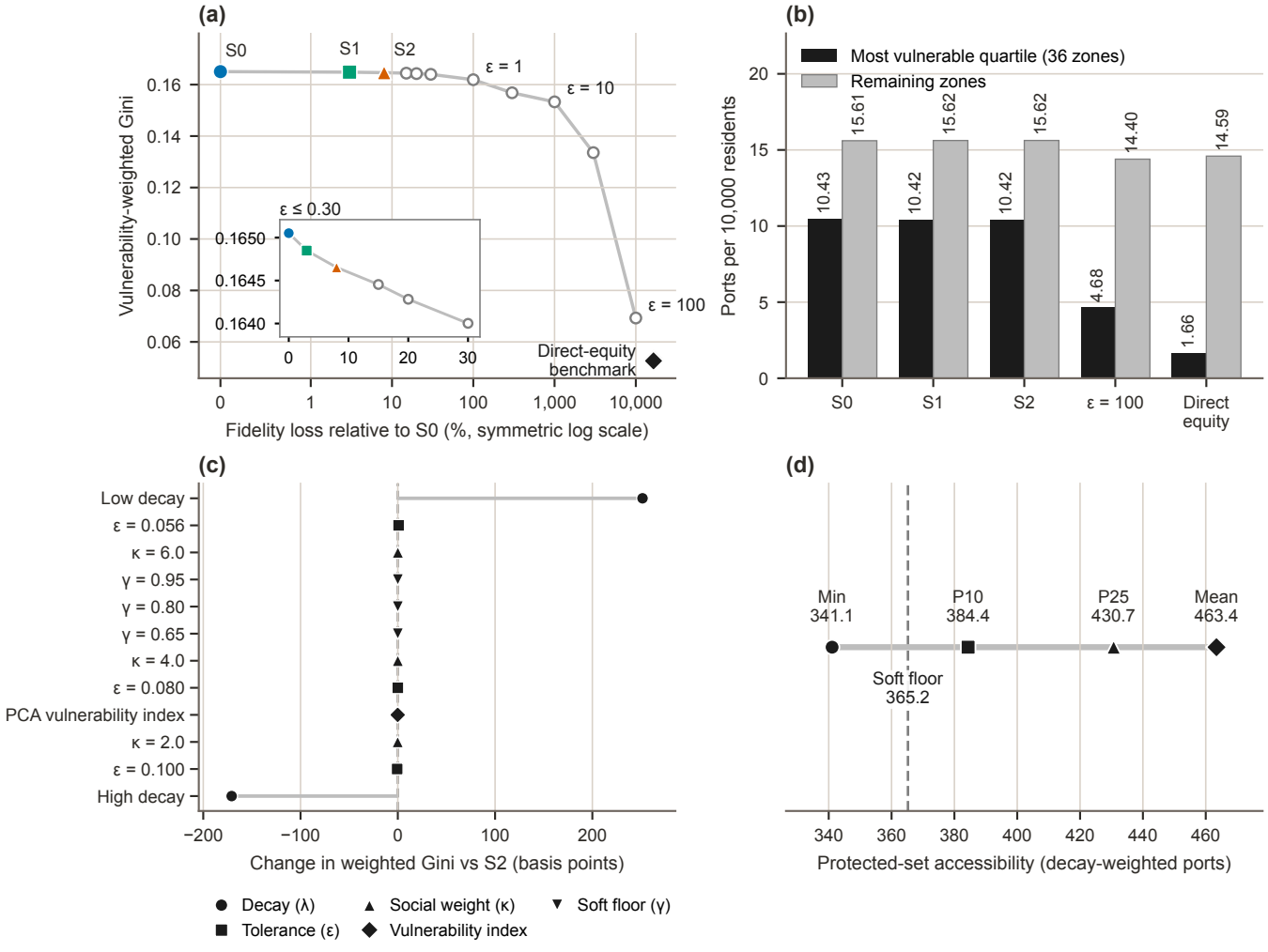
(a)
S0
S1
S2
ε = 1
ε = 10
ε = 100
ε ≤ 0.30
Direct-equity benchmark
Vulnerability-weighted Gini
Fidelity loss relative to S0 (%, symmetric log scale)
(b)
Most vulnerable quartile (36 zones)
Remaining zones
Ports per 10,000 residents
10.43
15.61
10.42
15.62
10.42
15.62
4.68
14.40
1.66
14.59
S0
S1
S2
ε = 100
Direct equity
(c)
Low decay
ε = 0.056
κ = 6.0
γ = 0.95
γ = 0.80
γ = 0.65
κ = 4.0
ε = 0.080
PCA vulnerability index
κ = 2.0
ε = 0.100
High decay
Change in weighted Gini vs S2 (basis points)
Decay (λ)
Social weight (κ)
Soft floor (γ)
Tolerance (ε)
Vulnerability index
(d)
Min 341.1
P10 384.4
P25 430.7
Mean 463.4
Soft floor 365.2
Protected-set accessibility (decay-weighted ports)